25900061 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCECT101 - DATA COMMUNICATION

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Define the formula for Shannon's capacity? [K] / [CO1]

   a). Capacity = 2 × bandwidth × $\log_2(L)$

   b). Capacity = bandwidth × $\log_2(1 + SNR)$

   c). Capacity = bandwidth / (1 + SNR)

   d). Capacity = bandwidth × L^2

2. What happens to the wavelength of sound waves as the frequency increases? [U] / [CO1]

   a). Wavelength increases

   b). Wavelength decreases

   c). Wavelength remains the same

   d). Wavelength becomes zero

3. In which type of data flow does data travel in only one direction? [U] / [CO1]

   a). Simplex

   b). Half-Duplex

   c). Full-Duplex

   d). None of the above

4. Recall that the Parabolic dish antennas are widely used in terrestrial microwave communication due to their ability to: [U] / [CO2]

   a). Minimize attenuation
   b). Transmit signals in all directions
   c). Focus signals into narrow beams
   d). Reduce power consumption

5. Identify the correct feature is commonly found in BNC Connectors [U] / [CO2]

   a). Screw-on locking mechanism
   b). Push-Pull locking mechanism
   c). Bayonet locking mechanism
   d). All of the above

6. The constellation diagram in QAM is used for to represent [K] / [CO3]

   a). the amplitude
   b). both amplitude and phase
   c). frequency
   d). phase

7. Provide an example of synchronous transmission from the options given. [U] / [CO3]

   a). USB (Universal Serial Bus)
   b). Email communication
   c). RS-232 serial communication
   d). Morse code transmission

8. Select the correct feature of serial communication: [U] / [CO3]

   a). Data is transmitted in parallel over multiple lines
   b). Data is transmitted bit by bit using one communication line
   c). Data transmission is limited to short distances only
   d). It uses a clock signal to synchronize data transmission

9. The technique that expands the bandwidth of a signal by replacing each data bit with n bits using a spreading code. [U] / [CO3]

   a). DSSS b). FHSS c). FDM d). WDM

10. The relationship between bit rate and baud rate is [U] / [CO3]

    a). Bit rate is always equal to baud rate
    b). Bit rate is always less than baud rate

c). Baud rate represents the number of signal units per second, while bit rate represents the number of bits per second

d). Bit rate and baud rate are unrelated

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Describe the relationship between bitrate and bandwidth in a communication system. [U] / [CO1]

12. Explain the different types of data representation in communication systems with examples. [U] / [CO1]

13. How to reduce the signal loss in coaxial cable? [U] / [CO2]

14. Describe the working of Adaptive Delta Modulation (ADM). How does it differ from standard Delta Modulation (DM)? [U] / [CO3]

15. Explain the need for Analog to Digital Conversion in communication systems. [U] / [CO3]

16. Explain the concept of multiplexing and its types. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Discuss the primary types of transmission impairments in communication system and explain their causes, effects on signal integrity, and common mitigation techniques. [U] / [CO1]

18. Describe the construction of a twisted pair cable,including the purpose of twisted the wires and the different types of shielding(UTP and STP) [U] / [CO2]

19. Discuss the advantage , disadvantage and applications of ASK and FSK. [U] / [CO3]

[10x2 = 20]